

Semester Examination – December - 2019

Sem.:-V

Marks: 60

Time:- 3 Hr.

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately Assume it and should mention it clearly

		(Marks)
Q1. A)	Explain data abstraction and different data models in DBMS	6M
B)	Explain different advantages of DBMS over file processing systems	6M
Q2. A)	From given tables, write relational algebra expressions for following branch (branch_name, branch_city, assets) customer (customer_name, customer_street, customer_city) account (account_number, branch_name, balance) loan (loan_number, branch_name, amount) depositor (customer_name, account_number) borrower (customer_name, loan_number)	1*5=6M
	a. Find all loans of over Rs.6000 b. Find the loan number for each loan of an amount greater than Rs.50000 c. Find the names of all customers who have a loan, an account, or both, from the bank d. Find the names of all customers who have a loan and an account at the bank e. Find names of customer who have an account in all branches located at Aurangabad. f. Find name of customer who lives in pune	
B)	Explain unary and binary operations in relational algebra.	6M
Q3. A)	Explain DDL, DML and DCL commands with suitable examples.	6M
B)	From given tables , write SQL queries for following Salesman(s_id, name, city, commission) Customer(c_id, c_name, city, grade, s_id) Orders(o_no, amount, date_of_order, customer_id)	1*5=6M
	a. Find out name of customer for each order b. Find out name of customer, who has placed 2nd highest amount of order c. Find out names of customers who have who have placed order of higher amount than Ram. d. Find out city wise count of customers. e. Find out names of customers who lives in same city as that of salesman f. Find c_id,s_id and date for each order.	

- Q4. A) Explain RAID in detail 6M
B) Explain normalization with different normal forms with suitable example. 6M
Q5. A) Explain ACID properties of transaction using suitable examples 6M
B) Explain different aggregate functions in SQL with suitable examples. 6M
Q6. A) Explain different concurrency control protocols 6M
B) Write short note on backup and recovery systems 6M

Paper End
